

Facial Emotion Detection and Recommendation using Deep learning

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Abstract— Facial emotion detection has emerged as a key focus in psychology, leveraging computer vision and machine learning to decode human emotions through facial expressions. By analyzing behaviors, actions, poses, and speech, it offers insights into individuals' emotional states. Deep learning models, particularly Convolutional Neural Networks (CNNs), have significantly improved accuracy in identifying emotions like anger, happiness, and sadness from facial expressions. This technology facilitates personalized content recommendations based on detected emotional states, enhancing human-machine interactions.

Facial emotions serve as a vital window into one's internal feelings and mental state, playing a crucial role in non-verbal communication. Despite various channels of expression, facial emotion detection stands out as the most utilized method for its ability to accurately reflect a person's emotional spectrum. Specialized computer programs, trained on annotated datasets, employ advanced mathematical techniques to decipher emotions, contributing to our understanding of human emotions and enhancing the reliability of emotion recognition systems. Through the integration of CNNs, these programs have become more precise and dependable, paving the way for more nuanced interpretations of emotional cues in human-machine interactions.

Keywords—Facial Emotion Detection, Convolutional Neural Network, Deep Learning, Machine Learning

I. INTRODUCTION

Facial expressions are really important in how we communicate without words. They quickly show various emotions, thoughts, and what we mean. Human faces can form numerous expressions, and these expressions are often universal, meaning they are recognized and understood across different cultures. An emotion is a feeling you have in your mind and body that's personal and unique to you. It includes how you act, what you do, what you think, and how you feel.

While wearable sensors can help identify emotions, using visual cues instead of physical connections is better because it's more flexible and significant [2]. People use facial

expressions to show how they feel, and it's a language everyone understands. Someone's expression can tell you a lot about what they're feeling.

Recent advancements in facial identification and feature extraction have made categorizing facial expressions and operating automated systems more complex. These improvements, driven by developments in machine learning and computer vision, have enhanced the accuracy of facial recognition algorithms, even in challenging conditions. In our project, we use a method based on Convolutional Neural Networks (CNN) to figure out different facial expressions. Our idea is to train the CNN to predict which emotion matches a face from a list of options: angry, happy, fearful, sad, disgusted, or neutral. Based on these identified facial sentiments our work recommends content according to it.

The paper is structured like this: first, we discuss previous research in the second section. Then, we explain our method in the next section. Finally, we wrap up with a short conclusion.

II. LITERATURE REVIEW

The authors studied a system for recognizing emotions on faces. They used electrooculography (EOG) signals to detect movements in the eyebrows and eyes. They figured out and showed various eye movements like looking up, down, right, left, blinking quickly, and making a face using symbols. The aim was to see if these movements could show different emotions on the face. [12]

In [11], the authors introduced a novel approach to face detection, which involves gathering and integrating data from thermal imaging and visible motion pictures. This method enabled a thorough exploration of the correlation between the intensity of deliberate facial expressions conveying emotions and the temperature of the facial surface. Additionally, in

Facial features in images were obtained by using Linear Discriminant Analysis (LDA), detecting facial landmarks after converting to grayscale, and then making conclusions based on these detections.

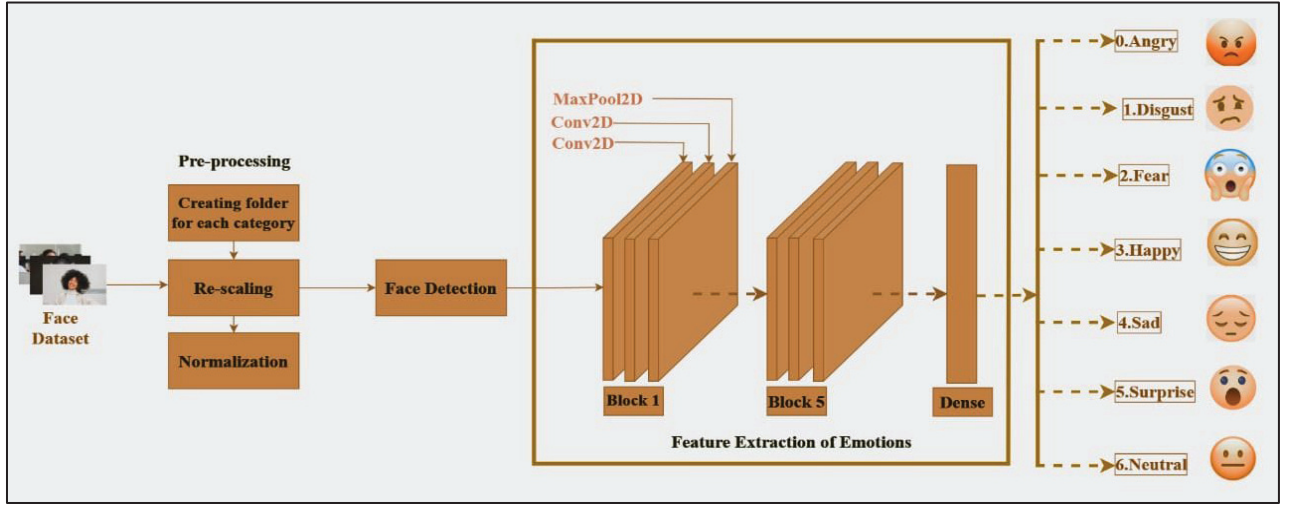


Fig. 1. Detection Architecture

They found people's emotional states by using a mix of K-Nearest Neighbors (KNN) and Hidden Markov models.

In a study mentioned in [14], scientists used KNN to identify facial expressions and got it right about 82.87% of the time. [15] employed CNN to explore emotion recognition by demonstrating the viability of identifying Facial Action Units (FAU). These components are included in the Facial Action Coding System (FACS), which identifies particular movements of facial muscles, helping to enhance detection techniques. One of the most basic ways people communicate is by expressing their feelings [1]. This happens without words and is important in our daily interactions. We call the process of figuring out how someone feels, especially by looking at their face, "emotion recognition." There have been many different ways tested to do this from face pictures, and quite a few of them have been pretty accurate.

Existing systems for recognizing user behavior are often complex in terms of time and storage. Leveraging the human face's significance, these systems employ face detection, feature extraction, and emotion recognition to identify user moods [17]. This facilitates automated tasks such as playing music and videos and providing motivational content, saving time and effort [18]. Convolutional neural networks and support vector machines are employed for emotion classification, validated through testing and training methods. Predicted emotions determine the applications offered, streamlining user interaction [20].

With the move toward the existing literature, the primary focus centers on crafting an emotion detection and recommendation system. The core of the system harnesses the power of Convolutional Neural Networks (CNNs), demonstrating exceptional proficiency in discerning facial emotions. Post-emotion detection, the cutting-edge recommendation system takes the spotlight. It operates by comprehending the user's emotional state, allowing the system to customize content recommendations in alignment with the user's feelings.

III. METHODOLOGY

In this part, we talk about the steps we took to create a model that can recognize facial emotions using the Convolutional Neural Network (CNN) algorithm. Our goal was to make a system that can accurately identify and predict

one of seven emotional states: Happy, Sad, Angry, Disgust, Surprise, Fear, and Neutral. The research involves implementing various deep-learning techniques to create an emotion recognition model. Facial emotion detection comprises a series of steps and follows a specific methodology for recognizing and classifying emotions from facial expressions. In the methodology section, there are two parts emotion detection & recommendation.

In this project, we use the CNN algorithm for our method. This emotion recognition system based on deep learning has two main parts:

- (1) CNN is trained using the FER-2013 database.
- (2) Seven types of emotions are detected.

A. Emotion Detection

- **FER-2013 Dataset:** The FER (Facial Expression Recognition) 2013 dataset is widely used in computer vision and emotion recognition. It's made to train and test programs that recognize facial expressions. The dataset includes black-and-white pictures of human faces, each marked with one of seven emotions. Here are some details about the FER 2013 dataset: Number of Images: There are 35,887 pictures in the dataset. Facial Expressions: The dataset shows seven different emotions: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

TABLE I. FER-2013 DATASET

Classification	Validation Data		Training	Dataset
	Public	Private	Data	Total
Angry	467	491	3995	4953
Disgust	56	55	436	547
Fear	496	528	4097	5121
Happy	895	879	7215	8989
Sadness	653	594	4830	6077
Surprise	415	416	3171	4002
Contempt	607	626	4965	6198
	3589	3589	28709	35887

- **Training Phase & Detection:** A Convolutional Neural Network (CNN) is a special kind of computer program made to understand and work with pictures and videos. It's good at tasks like figuring out what's

in a picture, finding objects in an image, and recognizing emotions on people's faces. Feature extraction involves the identification and extraction of relevant facial landmarks and regions. Commonly used landmarks include the eyes, nose, mouth, and various points on the face. These landmarks help the system focus on specific areas of the face for emotion analysis. In the feature extraction step, a CNN model comprises various layers, including convolutional, max pooling, and dense layers, which work together to extract different details from an input, such as a facial image. These layers collaborate to effectively analyze and process the data. The training process for the CNN model involves adjusting key parameters such as the learning rate, batch size, and the number of epochs. In this case, the model uses a learning rate of 0.005, a batch size of 64, and 60 epochs. The Conv2D and dense layers are especially crucial for the task of analyzing the extracted features and assigning emotional labels to the detected faces. Facial emotion detection system architecture integrates hardware and software, with deep learning pivotal in promptly and precisely analyzing emotional expressions.

- *Proposed Architecture:* The system architecture of facial emotion detection is a complex framework that combines various components to accurately identify and analyze the emotional expressions on a person's face. Here is an overview of the key elements of this architecture

Webcam testing architecture for facial emotion detection focuses on assessing the performance and accuracy of an emotion recognition system using a webcam as the input source. *This* architecture helps ensure that the system can effectively detect and classify emotions in real-time video streams. In this section, the primary aim is to use a webcam to identify facial expressions.

In which the webcam captures the facial condition of a particular user, according to that trained CNN model recognizes, *detects* & predicts the accurate emotion state.

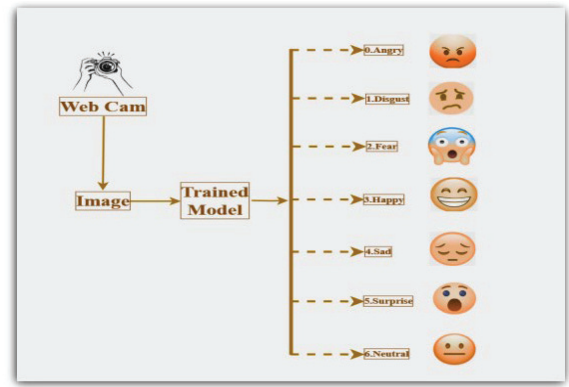


Fig. 2. Testing Architecture

The webcam testing architecture for facial emotion detection ensures that the system performs accurately and efficiently in real-time video streams. It also focuses on user interaction and adaptability to various conditions, ultimately leading to a robust and user-friendly solution for emotion recognition.

B. Recommendation

In the recommendation part, our system offers a broad method for suggesting motivational videos, focusing on building a strong model for this type of content.

We understand that videos can touch people's feelings, so we're putting a lot of effort into picking out motivational videos. Our main aim is to affect how users feel by showing them videos they can connect with. We want to make sure users feel good and motivated by giving them videos that match their emotions and what inspires them.

The recommendation system architecture begins with a user interface presented on a webpage, featuring language and speaker selection options tailored to enhance user experience. Users are empowered to choose their preferred language, with selections including popular languages such as Hindi and English.

Users can enhance their browsing experience by manually choosing from a diverse range of motivational speakers. Upon user interaction, the system employs language detection mechanisms to discern language preferences in cases where users do not explicitly specify.

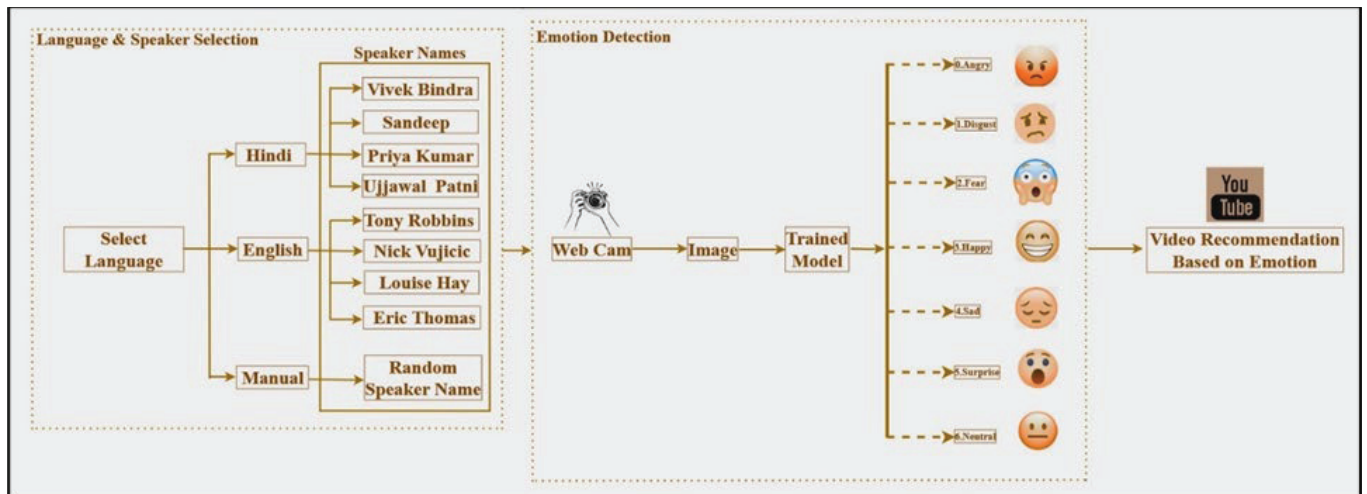


Fig. 3. System Architecture

Moreover, users have the option to make their experience better by picking particular motivational speakers. This ensures that the recommendations they get are more personal and match what they're interested in and inspired by. This method creates an exciting place where users can check out motivational stuff that fits with their language and what inspires them.

IV. RESULT & DISCUSSION

The following are the results obtained for video recommendation based on emotion detection.

Step: 1 Select the language

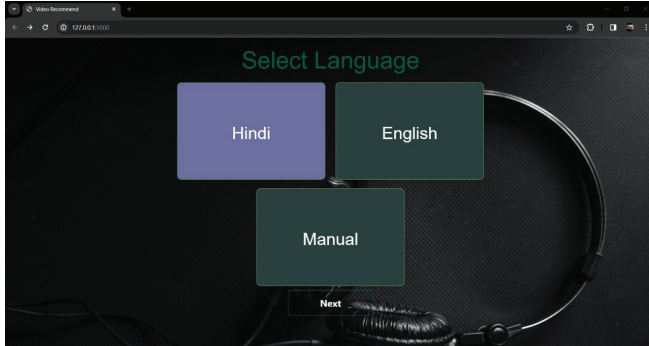


Fig. 4. Language Selection Interface

Our recommendation system initiates the process of suggesting motivational videos tailored to the user's mood. The user begins by specifying their preferred language, choosing between Hindi or English. Alternatively, they have the option to manually input their language preference. This personalized approach ensures that users receive content that resonates with them on an emotional and linguistic level. Additionally, our system continuously learns from user interactions to enhance recommendation accuracy and relevance over time.

Step: 2 Selection of Speakers

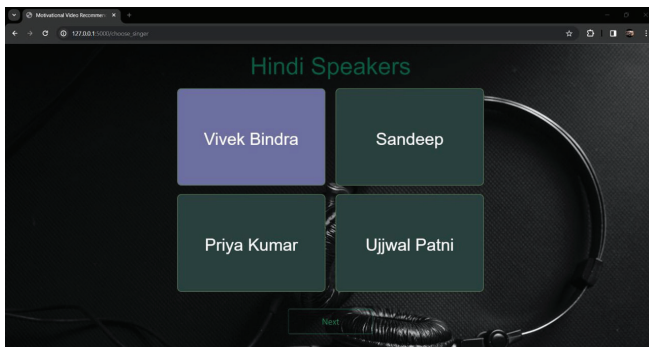


Fig 5(a). Hindi Speakers Selection Option

Upon selecting the Hindi language, the user will be presented with a range of options displayed on the screen. These options cater to different preferences, allowing users to make selections based on their tastes and interests. Whether it's genres, themes, or specific topics, users have the freedom to choose content that aligns with their preferences. This customizable experience ensures that users can engage with motivational videos that resonate with them on a deeper level. Additionally, our system continually adapts to user

interactions, refining recommendations to better suit each user's unique preferences and needs.

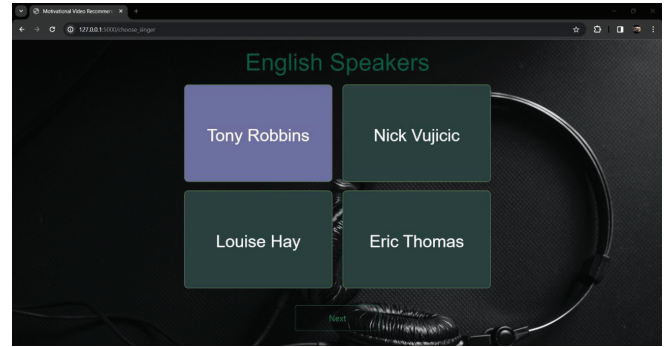


Fig 5(b). English Speakers Selection Option

If the user selects the English language, they will undergo a similar process. Upon choosing English, the system will present various options on the screen, allowing users to customize their experience based on their preferences. These options span across different categories, enabling users to select content that aligns with their interests and motivations. From genres to specific themes, users have the flexibility to tailor their choices according to their unique preferences. This personalized approach ensures that users receive recommendations that resonate with them emotionally and intellectually. Furthermore, our system continuously learns from user interactions to enhance the recommendation process, refining suggestions to better suit each user's preferences over time.

Step: 3 Manual Selection

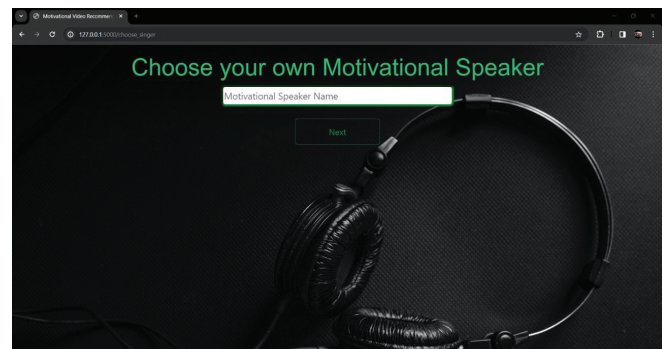


Fig. 5. (c). Manually Speakers Selection Option

If the user prefers a manual selection of speakers, they can simply input the name of the desired speaker. This option allows users to have direct control over their viewing experience, enabling them to access content from their favorite speakers or personalities. By entering the specific speaker's name, users can quickly discover and engage with motivational videos delivered by individuals they admire or resonate with. This hands-on approach empowers users to curate their viewing journey, ensuring they receive content that inspires and uplifts them. Additionally, our system facilitates seamless navigation and search functionality, making it easy for users to find the exact content they're looking for.

Step: 4 Emotion Detection

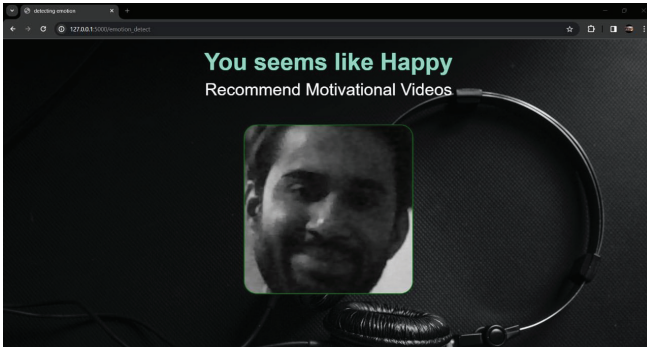


Fig 6(a). Emotion Detected as Happy

In the emotion detection phase, our system can identify a wide range of facial expressions, including happiness, fear, neutrality, disgust, and more.

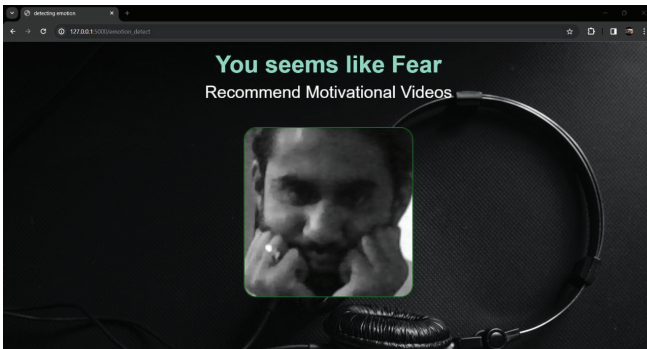


Fig. 6. (b). Emotion Detected as Fear

Using CNN algorithms, our system accurately detects and interprets the user's emotional state based on their facial cues. By analyzing various facial features and expressions, we ensure precise recognition of the user's emotions.

Step: 5 Video Recommendation on YouTube

The final stage of our process involves recommending motivational videos on YouTube based on the user's detected emotion.

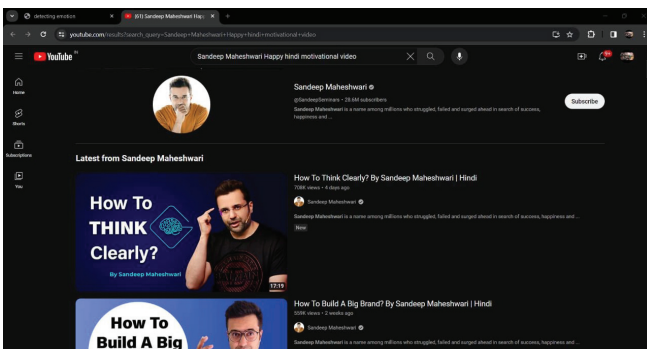


Fig. 7. Personalized YouTube Recommendation for Happy Emotions

Leveraging the insights gathered from emotion detection, our system curates a selection of videos that align with the user's current emotional state. Whether the user is feeling happy, anxious, neutral, or any other emotion, our recommendation engine ensures that they receive content that is relevant and uplifting.

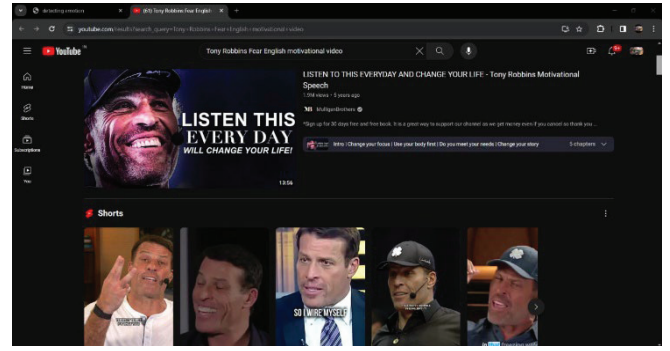


Fig. 8. Personalized YouTube Recommendation for Fear Emotions

By harnessing the power of machine learning algorithms, we continuously optimize our recommendations to provide a personalized viewing experience for each user. This tailored approach aims to inspire and empower users, guiding them toward content that resonates with their emotions and motivates them to achieve their goals.

V. CONCLUSION

In conclusion, facial emotions are pivotal in non-verbal communication, offering valuable insights into individuals' internal feelings. While recognizing diverse emotions accurately can be challenging, our study proposes a method using Convolutional Neural Networks (CNNs) to identify emotions like anger, disgust, happiness, fear, sadness, calmness, and surprise. However, beyond generalized facial emotion recognition, our research focuses on developing a system specifically tailored to recommending apps that align with users' emotional states, particularly emphasizing motivational video content. By harnessing the emotive power of videos, our system aims to curate personalized recommendations that uplift users' moods and actively shape their emotional experiences. This approach promises to enrich user engagement and foster a more fulfilling user experience through emotionally resonant content suggestions.

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